

MRIDUL

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EDUCATION

Bachelor of Technology in Computer Science Engineering

August 2022 – June 2026

[Government College Dharamshala](#) | Dharamshala

12th Class – Non-Medical

April 2021 – March 2022

[GSSS Daroh](#) GPA: 92.8%

EXPERIENCE

AI/ML Intern (WBL Scheme)

February 2025 – Present

[NIELIT Ropar](#) | Ropar, Punjab

- Working under the Work-Based Learning (WBL) Scheme, applying AI/ML concepts to real-world industry problems.
- Developing and deploying machine learning models as part of structured internship projects.
- Collaborating with mentors on end-to-end ML pipelines including data preprocessing, model training, and evaluation.

Google Student Ambassador

October 2025 – April 2026

[Google Gemini Student Program](#) | Government College Dharamshala

- Facilitated engagement campaigns and technical workshops to drive adoption of Google Gemini and Generative AI across campus.

Campus Leader

2026 – Present

[Hack Byte-Codex](#) | Government College Dharamshala

- Leading Hack Byte-Codex, a student-driven tech community focused on helping students learn, build skills, and connect with developers.
- Organizing workshops and webinars on emerging technologies including AI/ML, Web Development, and Open Source.
- Building a collaborative environment that bridges academics and industry-ready technical skills for college peers.

TRAINING & CERTIFICATIONS

Data Science Trainee

September 2025 – November 2025

[SmartED Innovation Ltd](#) | Online

- Completed comprehensive Data Science training, building end-to-end projects using Python (Pandas, NumPy) and Machine Learning algorithms.

Fundamentals of Data Curation using Python

June 2025 – July 2025

[NIELIT Ropar](#)

- Preprocessed and visualized multi-source data using Python (Pandas, NumPy, Seaborn) to optimize downstream ML/DL model training.

DSA using Python

October 2024 – Present

[mysirg.premium](#) | Online

- Completed hands-on DSA training implementing linear/non-linear data structures and optimizing algorithmic performance.

Complete Python Bootcamp – Zero to Hero

June 2024 – Present

[Udemy](#) | Online

- Mastered Python fundamentals and advanced concepts (OOP, Data Structures, File Handling) through practical coding challenges.

PROJECTS

AI-Powered Proctored Examination Platform | [Python](#) · [Flask](#) · [OpenCV](#) · [Media Pipe](#) · [Socket.IO](#) · [JavaScript](#)

- Developed a server-side AI proctoring system optimized for low-resource devices by shifting heavy computation to the backend.
- Built a Python/Flask backend with Media Pipe achieving 94% accuracy in detecting gaze deviation and impersonation.
- Created a thin-client interface using JavaScript and Socket.IO for smooth video streaming and instant lag-free alerts.

Digital Gram Samadhan Portal | [Live Demo](#)

[Python](#) · [Flask](#) · [MongoDB](#) · [Sarvam AI](#) · [Socket.IO](#) · [JavaScript](#)

- Engineered a multimodal rural e-governance platform featuring AI-powered voice-to-text grievance registration, supporting 9 regional languages to overcome user literacy barriers.
- Developed a secure Python/Flask backend and scalable MongoDB NoSQL database, implementing strict Role-Based Access Control (RBAC) and automated 7-day SLA escalation for delayed complaints.
- Built a responsive, mobile-first interface using JavaScript and Jinja2, integrating Socket.IO for real-time dashboard updates and Cloudinary for secure, optimized media storage

NIELIT Smart Entry System | [Live Demo](#)

[Python](#) · [Flask](#) · [Computer Vision \(QR\)](#) · [Docker](#) · [JavaScript](#)

- Engineered a secure QR-code-based access control and visitor management system for NIELIT Ropar to streamline facility entry and eliminate manual logging bottlenecks.
- Developed a robust Python/Flask backend with strict role-based access isolation, featuring real-time IN/OUT tracking, manual visitor ID capture, and automated day-wise CSV data exports.
- Built a responsive, role-separated scanner interface integrating live camera feeds for instant QR verification, and deployed the production-ready application utilizing Docker and Gunicorn.

Multi-Crop Disease Detection and Advisory Platform | [Python](#) · [Flask](#) · [TensorFlow](#) · [Efficient Net B0–B3](#) · [Scikit-learn](#)

- Built a scalable platform covering 8 regional crops; trained Efficient Net (B0/B2/B3) models reaching 80–90% accuracy.
- Handled class imbalance using class weights; developed a lightweight Flask + Tailwind CSS + JS interface for instant diagnoses and advice.

Sales Forecasting for a Retail Chain | | [Live Demo](#)

[Python](#) · [SQL](#) · [Regression](#) · [EDA](#) · [Matplotlib](#) · [HTML/CSS](#) · [JavaScript](#)

- Developed a forecasting system with SQL data extraction, Python EDA for trend/seasonality analysis, and ML models (Linear Regression, Random Forest, XGBoost).
- Built an interactive Flask dashboard visualizing insights, model metrics, and actionable sales forecasts for stakeholders.

SKILLS

Languages: Python · C++ · Java · MySQL · HTML/CSS · JavaScript

Frameworks: Flask · TensorFlow · Tailwind CSS · Streamlit

Developer Tools: Git · Hugging Face · Render · VS Code · PyCharm · Kimi · Github Copilot · LLM Models · Claude AI / Z.AI

Libraries: Pandas · NumPy · Matplotlib · Seaborn · Scikit-learn · OpenCV · MediaPipe